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Esta é uma cópia de uma conversa entre ChatGPT e Anônimo.

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If $Z_1, \dots, Z_n$ is an ergodic sample from a cdf F with density f , it is well known that there are no unbiased estimators of f . In Proposition A.1 from Kokoszka et al. [1] it is shown that, for f in a suitable class $\mathbb{D}$, it holds that $\Lambda(f) := -\log f \circ Q$ lies in $L^2[0,1]$, where Q is the generalized inverse (aka quantile function) of F .

Consider the kernel density estimator $\hat{f}$ of f defined as

$$\hat{f}(x) = \frac{1}{nh} \sum_{t=1}^n K\left(\frac{x - Z_t}{h}\right)$$

where K is some probability density function with $\int x K(x) dx = 0$. Is it possible that $\Lambda(\hat{f})$ is unbiased for $\Lambda(f)$?

[1] Piotr Kokoszka, Hong Miao, Alexander Petersen, Han Lin Shang, Forecasting of density functions with an application to cross-sectional and intraday returns, International Journal of Forecasting, Volume 35, Issue 4, 2019, Pages 1304-1317, ISSN 0169-2070, <https://doi.org/10.1016/j.ijforecast.2019.05.007>.

I'm checking two layers here: first the simpler pointwise transform $-\log \hat{f}(x)$, then the harder composition with the quantile map. The fixed- x case already suggests the answer is essentially no, except in degenerate edge cases.

Thought for 2m 19s >

In the usual statistical sense, **no**: $\Lambda(\hat{f})$ is not an unbiased estimator of $\Lambda(f)$ on any nontrivial class of densities.

The cleanest way to see this is to start with the easier object

$$\tilde{\Lambda}_n(u) := -\log \hat{f}(Q(u)),$$

where Q is the **true** quantile function. If even this is not unbiased, then the fully plug-in version

$$\Lambda(\hat{f})(u) = -\log \hat{f}(\hat{Q}(u))$$

has even less chance, because $\hat{Q}$ introduces an extra layer of randomness.

1) Expectation of the KDE

If (Z_t) is stationary ergodic with marginal density f , then for each fixed x ,

$$\mathbb{E}[\hat{f}(x)] = \frac{1}{nh} \sum_{t=1}^n \mathbb{E}\left[K\left(\frac{x - Z_t}{h}\right)\right] = \frac{1}{h} \mathbb{E}\left[K\left(\frac{x - Z_1}{h}\right)\right] = (K_h * f)(x),$$

where $K_h(x) = h^{-1}K(x/h)$.

So the KDE is centered at the smoothed density $K_h * f$, not at f .

2) The log transform adds its own bias

Since $-\log$ is strictly convex on $(0, \infty)$, Jensen gives, for every u such that $\hat{f}(Q(u)) > 0$ a.s.,

$$\mathbb{E}[-\log \hat{f}(Q(u))] > -\log \mathbb{E}[\hat{f}(Q(u))] = -\log((K_h * f)(Q(u))),$$

unless $\hat{f}(Q(u))$ is almost surely constant, which it is not in any nondegenerate problem.

Therefore, if $\tilde{\Lambda}_n$ were unbiased, we would need

$$-\log f(Q(u)) = \mathbb{E}[-\log \hat{f}(Q(u))] > -\log((K_h * f)(Q(u))),$$

hence

$$(K_h * f)(Q(u)) > f(Q(u)).$$

That inequality would have to hold for all u and all f in your class. But it is false in general.

3) Why this fails generically

Take a smooth density f with a strict interior mode at x_0 , so $f''(x_0) < 0$. Under the standard KDE assumption that K has finite second moment $\mu_2(K)$,

$$(K_h * f)(x_0) = f(x_0) + \frac{h^2}{2} \mu_2(K) f''(x_0) + o(h^2),$$

so for small h ,

$$(K_h * f)(x_0) < f(x_0).$$

Let $u_0 = F(x_0)$. Then $Q(u_0) = x_0$, and the condition above would require

$$(K_h * f)(x_0) > f(x_0),$$

a contradiction.

So even the simpler estimator $-\log \hat{f}(Q(\cdot))$ cannot be unbiased for $-\log f(Q(\cdot))$ on any reasonable class containing such densities.

4) What about $-\log \hat{f}(\hat{Q}(u))$?

That is the natural interpretation of $\Lambda(\hat{f})$, since $\Lambda(g) = -\log g \circ Q_g$ uses the quantile of the same density g .

But then things get worse, not better. Now the bias comes from three sources:

- **smoothing bias** of $\hat{f}$: $(K_h * f) - f$,
- **nonlinearity bias** from $-\log$,
- **plug-in quantile bias** from replacing Q by $\hat{Q}$.

Under standard smoothness/mixing conditions, a delta-method expansion gives, schematically,

$$\mathbb{E}[\Lambda(\hat{f})(u)] - \Lambda(f)(u) = -\frac{((K_h * f) - f)(Q(u))}{f(Q(u))} + \frac{\text{Var}(\hat{f}(Q(u)))}{2f(Q(u))^2} + \text{terms from } \hat{Q} - Q +$$

so the bias is generically nonzero.

Bottom line

If by “unbiased” you mean

$$\mathbb{E}_f[\Lambda(\hat{f})(u)] = \Lambda(f)(u) \quad \text{for a.e. } u \in [0, 1], \forall f \in \mathbb{D},$$

then the answer is **no**.

At most, for a specially chosen single f , single u , and special h , you could get an **accidental finite-sample cancellation** of biases. But that is not an unbiased estimator in any meaningful classwise sense

If you want, I can also derive the leading asymptotic bias of

$$\Lambda(\hat{f})(u) = -\log \hat{f}(\hat{Q}(u))$$

more explicitly, in terms of f , f' , f'' , K , h , and the dependence structure of the ergodic sample.